

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
6	(a)	(i)	Rms pd is required or $V_{\text{rms}} = \frac{V}{\sqrt{2}}$ (1) Substitution into $P = \frac{V^2}{R}$ or equivalent e.g. calculating I , $P = IV$ (1)	2			2		
		(ii)	$\frac{18}{5} = 3.6$ squares seen or implied (1) $T = \frac{1}{f}$ used (0.0143 or 14.3 ms) (1) Dividing period by 2 ms (7.1 squares) (1) Period correct in diagram (ecf) (1) accept any phase, Amplitude correct in diagram (ecf) (1) at least one cycle	1	1 1 1 1		5	2	

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	(b)	(i)	Reactance of inductor equal and opposite to capacitance (or pds equal and opposite) (1) Minimum impedance (or all pd across resistor) (1) Maximum current (1)	3			3		
		(ii)	Algebra leading to $f = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$ (1) Answer = 130 kHz (1)		2		2	2	
		(iii)	$X_L = 65\,000\, [\Omega]$ or $X_C = 10\,500\, [\Omega]$ (1) Substitution into $Z = \sqrt{(X_L - X_C)^2 + R^2} = 55\,000\, \Omega$ (1) Use of $I = \frac{V}{Z}$ leading to Answer = 91 μA (ecf) (1)		3		3	3	
		(iv)	Smallest capacitor used (or trial and error) (1) Q factor calculated (533) (or other valid method of getting V) (1) Pd across C or L = $Q \times 5$ (2 700 V) (1) Peak pd this is $\times \sqrt{2}$ (3 800 V) (1) Student is wrong, it's considerably larger (valid conclusion) (1)			5	5	3	
			Question 6 total	6	9	5	20	10	0